Lenses and image formation

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Use ray directions to explain converging and diverging lens behavior.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 6, Day 11.*
Observed frequency is higher than 500 Hz because arriving wavefronts are closer together in time.

02. Worked example: explain the model
Real inverted image between f and 2 f on opposite side, smaller than the object.

03. Read and interpret the model
Converging rays toward far focus; diverging rays spread with back extensions to near focus.

04. Apply the physics
Inverted real image between f and 2 f for ideal converging lens.

05. Explain your reasoning
Image character depends object location; inside f emerging rays diverge and back extensions form virtual image.

06. Review: connect an earlier idea *Review: Unit 6, Day 2.*
6 cycles; frequency 1/4 Hz ≈ 0.25 Hz.

07. Exit check
Diverging.

08. STRETCH: test the model
They diverge; backward extensions give an upright magnified virtual image on the object's side.

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01. Warm-up: bring back an earlier idea *Review: Unit 6, Day 11.*
$f_{obs} \approx 523.1$ Hz, higher than emitted frequency.

02. Worked example: explain the model
Real inverted image between f and 2 f on opposite side, smaller than the object.

03. Read and interpret the model
Converging rays toward far focus; diverging rays spread with back extensions to near focus.

04. Apply the physics
d_i 15 cm; m=-0.5, inverted half size.

05. Explain your reasoning
Image character depends object location; inside f emerging rays diverge and back extensions form virtual image.

06. Review: connect an earlier idea *Review: Unit 6, Day 2.*
$\lambda = 0.4$ m; $f = 1/T \approx 16.67$ Hz.

07. Exit check
d_i=-10 cm; m=+2; upright virtual enlarged image.

08. STRETCH: test the model
d_i = -7.5 cm; m = +0.25. The image is virtual, upright and one-quarter the object's height, on the object side of the lens.